

Curriculum Vitae

Sep 2026

PERSONAL INFO

NAME: Boyuan Zhao
ADDRESS: BICMR, Peking University, Beijing, China 100871
PHONE: +44 7514180075, +86 18910504316
EMAIL: boyuan.zhao@pku.edu.cn
NATIONALITY: China
[PERSONAL WEBPAGE](#)

EDUCATION

10/2021- 09/2025	PHD IN MATHEMATICS, University of St Andrews, Supervised by Mike Todd and Kenneth Falconer Dissertation: <i>Exponential mixing and limit theorems in dynamical systems.</i>
09/2020-08/2021 Distinction (GPA 17.3/20)	MSc Mathematics University of St Andrews Dissertation: <i>On rotation number and unique minimal sets of circle homeomorphisms.</i> Advisor: Mike Todd
09/2017-08/2020 First Class Honours	BSc Mathematics with Modern Languages University College London

EMPLOYMENT

03/2026-	Postdoc - Beijing International Centre for Mathematical Researches (BICMR), Peking University, China
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RESEARCH INTERESTS

I am interested in various topics and problems in ergodic theory and dynamical systems. More precisely,

- Countable Markov shifts and Strong positive recurrence theory,
- Dynamical Borel-Cantelli lemmas and one-dimensional dynamics,
- Extreme Value Theory, Recurrence events, Waiting time distributions,
- Multifractal spectrum for maps modelled by countable Markov shifts.

RESEARCH OUTPUT

- (with Godofredo Iommi and Mike Todd) *Convergence of the Birkhoff spectrum for nonintegrable observables*, (2025) preprint, [ARXIV](#) (16 pages).
- *Almost sure convergence of cover times for ψ -mixing systems*, accepted by Ergodic Theory Dynam. Systems. (2025) [ARXIV](#).
- (with Mike Todd) *Countable Markov Chain with Exponential Mixing*. (2024) preprint, [ARXIV](#) (24 pages).
- *Closest distance between iterates of typical points*. Discrete Contin. Dyn. Syst. 44(8), (2024) 2252-2279. [PDF](#)

TEACHING EXPERIENCE

I am teaching Linear Algebra (C) at Peking University during the Fall semester 2026.

For each semester during my PhD, I have been a tutor and an example- class demonstrator for the following modules:

- MT1002 Mathematics

- MT1003 Pure and Applied Mathematics
- MT2501 Linear Algebra
- MT2502 Analysis

CONFERENCES ATTENDED AND TALKS

Aug 2026	Dynamical systems seminar, @SIMIS (Fudan University) <i>Talk title: Convergence rate of the Birkhoff spectrum for non-integrable observables.</i>
Mar 2026	Frontiers in Dynamical Systems,
Jun 2025	BMC & BAMC 2025, @ University of Exeter.
Jun 2025	Beyond Uniform Hyperbolicity II, ICTP
Mar 2025	Scottish Dynamics Network meeting, @ Herriot-Watt University <i>Talk title: Almost sure cover times in dynamical systems.</i>
Jul 2024	Workshop on Dynamical Systems: Geometric and Statistical Properties University of Warwick
Jun 2024	BMC 2024, @ University of Manchester <i>Talk title: Countable Markov shifts with exponential mixing.</i>
Dec/2023	Workshop on Probabilistic Methods in Dynamics @ Universidade do Porto <i>Talk title: Maximal length of subsequence matching and Renyi entropy.</i>

I have also given four talks at the University of St Andrews analysis seminar. See [GROUP PAGE](#) for titles and abstract.

SCHOLARSHIP AND FUNDING

10/2021-10/2025 China Scholarship Council Programme

OTHER INTERESTS AND SKILLS

I can speak Spanish at B2 level and Portuguese at B1.

I am an Irish & Scottish folk musician. [Video](#). I have been the vice president of the University of St Andrews folk and traditional music society, and the coordinator of our performance/ceilidh band.

I have competed 400m for University of St Andrews in the SSS indoor championships. I am now a distance/trail runner.